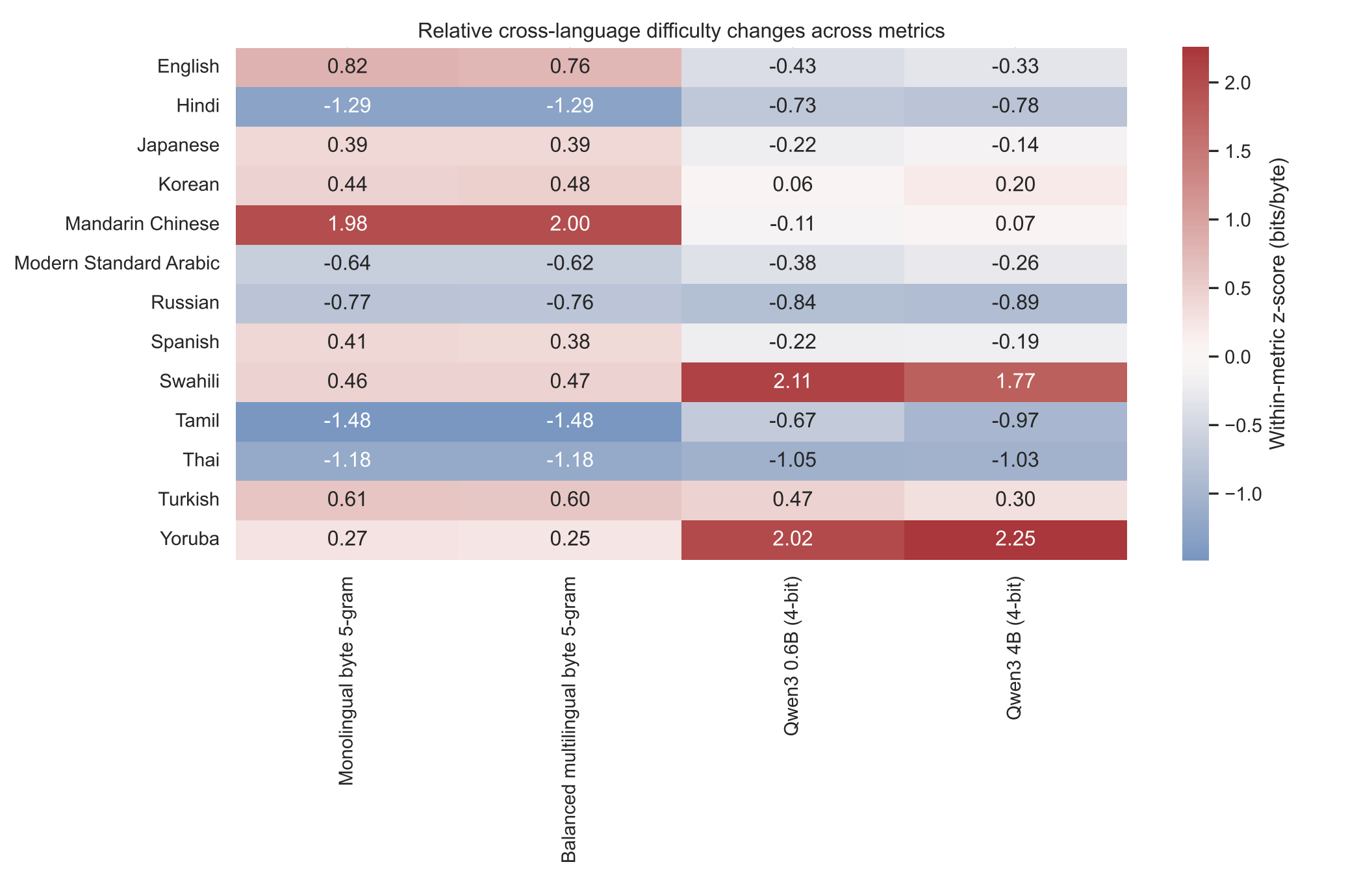
Relative cross-language difficulty changes across metrics
English 0.82 0.76 -0.43 -0.33
Hindi -1.29 -1.29 -0.73 -0.78
Japanese 0.39 0.39 -0.22 -0.14
Korean 0.44 0.48 0.06 0.20
Mandarin Chinese 1.98 2.00 -0.11 0.07
Modern Standard Arabic -0.64 -0.62 -0.38 -0.26
Russian -0.77 -0.76 -0.84 -0.89
Spanish 0.41 0.38 -0.22 -0.19
Swahili 0.46 0.47 2.11 1.77
Tamil -1.48 -1.48 -0.67 -0.97
Thai -1.18 -1.18 -1.05 -1.03
Turkish 0.61 0.60 0.47 0.30
Yoruba 0.27 0.25 2.02 2.25
Monolingual byte 5-gram
Balanced multilingual byte 5-gram
Qwen3 0.6B (4-bit)
Qwen3 4B (4-bit)
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
Within-metric z-score (bits/byte)